

Limiting & Excess Reagent

1. How many grams of water can be produced at most when 4 g of hydrogen and 8 g of oxygen are reacted?
2. $3,01 \cdot 10^{23}$ molecules of nitrogen and 3,36 L of hydrogen at STP are reacted. How many grams of ammonia are produced at most?
3. 18 g of carbon and 8 g of oxygen are reacted to form carbon dioxide. Which element and how many grams remain behind?
- *4. Equal weights of calcium and oxygen gas are reacted in a closed container to give 112 g of calcium oxide. Which element and how many grams will remain behind?
- *5. Equal volumes of sulphur dioxide and oxygen gases are reacted to give 5 liters of Sulphur trioxide at the same conditions. Which one remains unreacted and how many liters?

Problems Involving Impure Substances

1. How many grams of carbon dioxide are released when a 53 g of sodium carbonate sample that is 20 % reacts completely with hydrochloric acid ?
2. A 25 gram sample of iron can react completely with 200 g of 14,6 % hydrochloric acid solution by mass. Calculate the mass percentage of iron in the sample.
3. The reaction of 6 mol of hydrogen with 1,5 mol of nitrogen produces 2,4 mol of ammonia. What is the percent yield of the reaction ?
- *4. How many grams of magnesium sample that is 80 % pure, are needed to produce 11,2 L of hydrogen gas at STP from the reaction with hydrochloric acid ?